RECOVERING AN OBJECT AS THE DERIVED LIMIT OF ITS TRUNCATIONS FROM PER-OBJECT VANISHING BOUNDS

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file sites-cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof is the author's own argument: it supplies the explicit function making the companion criterion applicable and names that criterion. The pool line range for this proof stopped one line short of the closing \end{proof}; the span was aligned to the source (5411-5414) and the card moved to r002.

Editorial note. Every labelled reference occurring inside the counted spans is carried in assembly.external_references as a non-load-bearing navigation pointer (same-chapter ones at their definition site, cross-chapter ones at the invocation inside the counted span), so no load-bearing mathematics is replaced by a stub. No bibliography citation occurs inside the counted spans.

Editorial note. The checked chapter list for same-name collisions is the set of stacks chapters staged in this repository: chow, cohomology, divisors, etale-cohomology, flat, more-algebra, more-morphisms, morphisms, spaces-more-morphisms, varieties together with this card's chapter sites-cohomology.

Editorial note. The preamble is the chapter's own retained minimal preamble (sources/stacks-sites-cohomology/retained-preamble.tex), a verbatim excerpt of the native preamble: the Xy-pic setup and the native theorem declarations with the mathematical macros.

Allowed standard prerequisites (editorial).

- Derived limit of the system of truncations and the canonical map into it. Same chapter together with the chapter Derived Categories, the canonical map from an object to the derived limit of its truncations. Supplies the map whose being an isomorphism is asserted.
- Companion criterion with a per-object bounding function. Same chapter, the companion lemma giving the same conclusion from a function p(V, -) and a cofinal system of coverings for each object of the covering family. Supplies the result applied in the proof with the explicit function below.

Source reference. Original remark 0H72: sources/stacks-sites-cohomology/pinned-original-source.tex, source lines 5337–5337. Navigation pointer invoked at proof line 5337; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original lemma 0D6M: sources/stacks-sites-cohomology/pinned-original-source.tex, source lines 5320–5320. Navigation pointer invoked at proof line 5412; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Date: stacks-project:sites-cohomology.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

Lemma 0.1. *Let $(\mathcal{C}, \mathcal{O})$ be a ringed site. Let $E \in D(\mathcal{O})$. Let $\mathcal{B} \subset \mathrm{Ob}(\mathcal{C})$ be a subset. Assume*

(1) *every object of $\mathcal{C}$ has a covering whose members are elements of $\mathcal{B}$, and*
(2) *for every $V \in \mathcal{B}$ there exist an integer $d_V \geq 0$ and a cofinal system Cov_V of coverings of V such that*

$$H^p(V_i, H^q(E)) = 0 \text{ for } \{V_i \to V\} \in Cov_V,\ p > d_V,\ \text{and } q < 0$$

Then the map $E \to R\lim \tau_{\geq -n} E$ of Derived Categories, Remark 0H72 is an isomorphism in $D(\mathcal{O})$.

Proof. This follows from Lemma 0D6M with $p(V, m) = d_V + \max(0, m)$. □